

Too many columns. One picture

Dimensionality Reduction

Introduction to Unsupervised Learning in Classical Machine Learning



Hamid Bekamiri

Session 6 · BDS M1

What is Machine Learning?



Supervised ML

Algorithms learn from examples with known answers (labels). We provide the outcomes, and the model predicts outcomes for new, unseen data.



Unsupervised ML

No target variable to predict. Discovers hidden patterns in data without being told what to look for, revealing the "unknown unknowns."

The Fundamental Split in Learning

All machine learning boils down to one key question: **Do you have the "right answers" to learn from?**

The core distinction comes down to labels.



With Labels (Supervised)

Learning with a teacher. We have "right answers" available. The model learns to categorize or predict some value based on known target outcomes. (e.g., classifying emails as spam or not spam).



Without Labels (Unsupervised)

Learning through exploration. The model learns general patterns in the data, discovers how to represent data in a compressed way, or identifies related observations without any guidance or predefined answers.

Labeled vs. Unlabeled Datasets

Concrete examples of how data structure determines the machine learning approach

SUPERVISED LEARNING

Labeled Dataset Example

Features are paired with target labels ("ground truth").

	STATUS	SENDER	SUBJECT	SNIPPET	DATE
	SPAM	noreply@lottery.com	YOU WON \$1,000,000!	"Congratulations! Click here to claim your..."	Oct 27
	NOT SPAM	sarah.jones@corp.net	Project Update Meeting	"Team meeting on Monday at 10 AM to discuss..."	Oct 27
	SPAM	service@bank-alert.co	URGENT: Account Suspended	"Your account is temporarily locked. Validate now..."	Oct 26
	NOT SPAM	newsletter@designweekly.com	Design Inspiration	"The latest articles, tips, and resources..."	Oct 26
	NOT SPAM	j.doe@university.edu	Research Paper Feedback	"Hi Alex, attached are my comments on your..."	Oct 25

Email Subject (Feature)	Sender (Feature)	Label (Target)
Claim your free prize now!	promo@unknown.com	Spam
Quarterly Budget Meeting	boss@company.com	Not Spam
Your invoice #10492	billing@service.io	Not Spam

UNSUPERVISED LEARNING

Unlabeled Dataset Example

Raw input features without predefined outcomes or labels.



Customer ID	Annual Spend (\$)	Visit Frequency
CUST-8021	\$12,450	45 / year
CUST-8022	\$1,200	3 / year
CUST-8023	\$8,900	38 / year

Classical Machine Learning Landscape

Supervised Learning

Task Driven • Pre-Categorized Data

Classification

Divide the data into categories (e.g., separating socks by color).

→ Ex: Identity Fraud Detection

Regression

Predict continuous values (e.g., organizing ties by their length).

→ Ex: Market Forecasting

Objective: Predictions & Predictive Models

Unsupervised Learning

Data Driven • Unlabelled Data

Dimensionality Reduction

Discover wider dependencies in complex data.

→ Ex: Big Data Visualization

Clustering

Divide data by inherent similarity.

→ Ex: Targeted Marketing

Association

Identify underlying sequences or rules.

→ Ex: Customer Recommendation

Objective: Pattern & Structure Recognition

CHALLENGE 1

Customer Churn Dataset

Customer	Monthly Spend	Complaints	Contract	Churn
C101	\$95	4	Monthly	Yes
C102	\$42	0	Annual	No
C103	\$78	2	Monthly	Yes

CHALLENGE 2

Spotify/Music Discovery

Song	Tempo	Energy	Danceability	Acousticness
Song A	128	0.91	0.84	0.05
Song B	72	0.31	0.42	0.82
Song C	118	0.79	0.76	0.12

CHALLENGE 3

Customer Segmentation

Customer	Annual Spend	Visits/Year	Avg. Order
C201	\$12,500	48	\$260
C202	\$1,100	4	\$275
C203	\$7,800	35	\$223

CHALLENGE 4

Predict House Prices

Size (m²)	Bedrooms	Age	Distance to Center	Price (DKK)
85	2	25	8 km	2,100,000
140	4	10	4 km	3,750,000
190	5	3	12 km	4,200,000

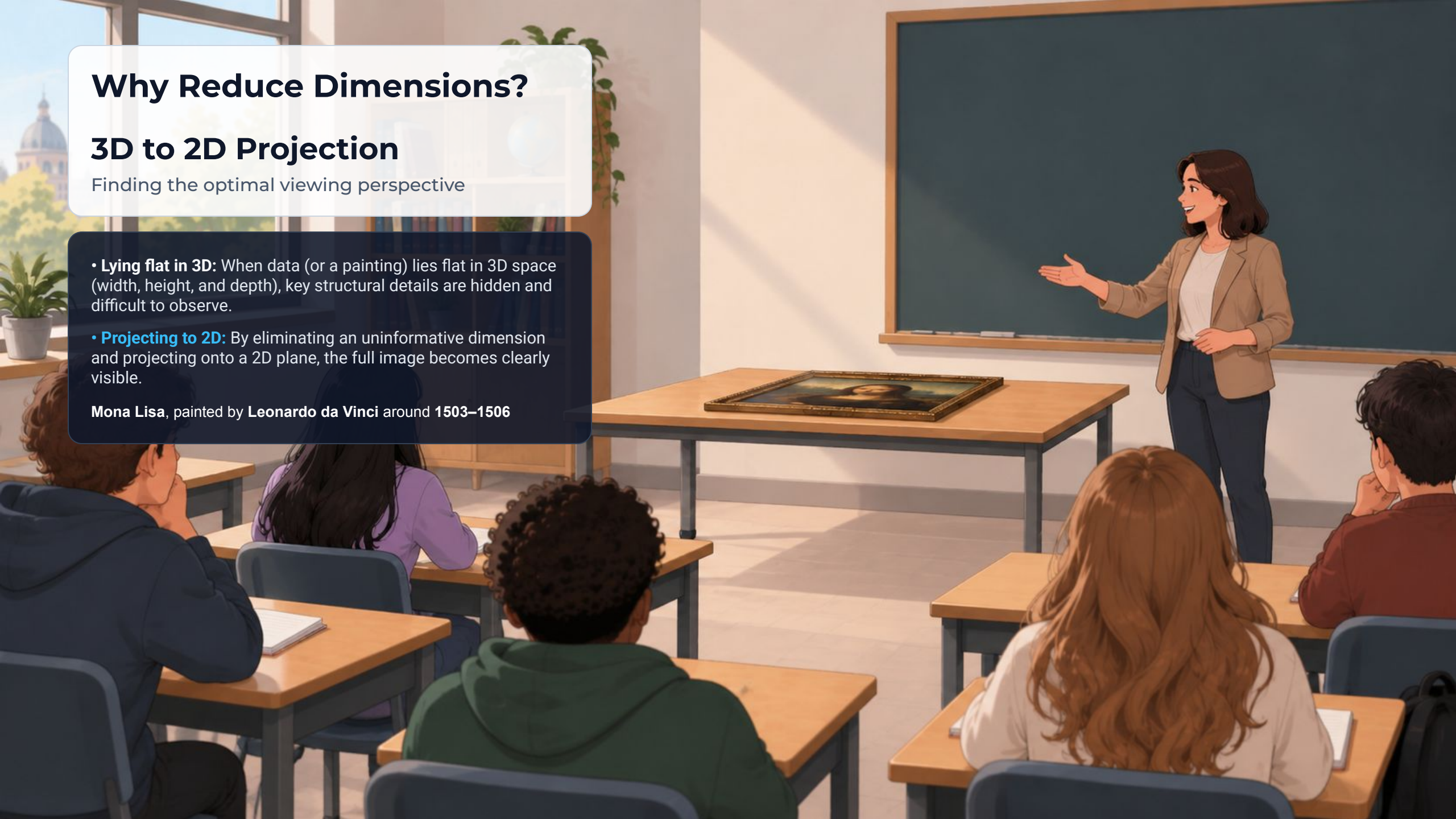
Why Reduce Dimensions?

3D to 2D Projection

Finding the optimal viewing perspective

- **Lying flat in 3D:** When data (or a painting) lies flat in 3D space (width, height, and depth), key structural details are hidden and difficult to observe.
- **Projecting to 2D:** By eliminating an uninformative dimension and projecting onto a 2D plane, the full image becomes clearly visible.

Mona Lisa, painted by **Leonardo da Vinci** around **1503–1506**



Why Reduce Dimensions?

3D to 2D Projection

Finding the optimal viewing perspective

- **Lying flat in 3D:** When data (or a painting) lies flat in 3D space (width, height, and depth), key structural details are hidden and difficult to observe.
- **Projecting to 2D:** By eliminating an uninformative dimension and projecting onto a 2D plane, the full image becomes clearly visible.

Mona Lisa, painted by **Leonardo da Vinci** around **1503–1506**



Dimensionality Reduction Methods

DIMENSIONALITY REDUCTION

LINEAR METHODS

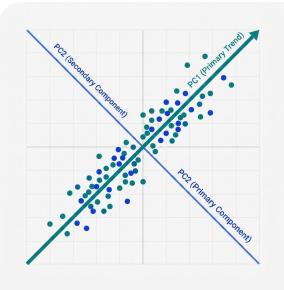
Straight-line structure preservation

NON-LINEAR METHODS

Curved & complex manifold structure

PCA

MAX VARIANCE



Finds orthogonal axes of max variance.

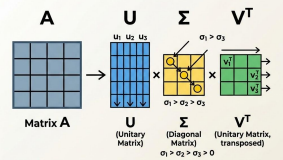
WHEN TO USE
High-dim feature reduction.

EXAMPLE
Customer segmentation.

SVD

MATRIX FACTOR

MATRIX DECOMPOSITION
($A = U \times \Sigma \times V^T$)



$A (m \times n) = U (m \times m) \times \Sigma (m \times n) \times V^T (n \times n)$

Factorizes data into linear vectors.

WHEN TO USE
Relational matrix factors.

EXAMPLE
Recommender systems.

NMF

ADDITIVE PARTS

ADDITIVE BLOCK COMPONENTS
BUILDING A LARGER STRUCTURE (NON-NEGATIVE VALUES)



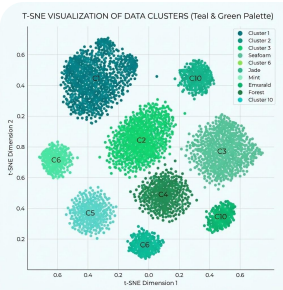
Creates interpretable positive parts.

WHEN TO USE
Additive positive features.

EXAMPLE
Topic modeling & audio.

t-SNE

LOCAL CLUSTERS



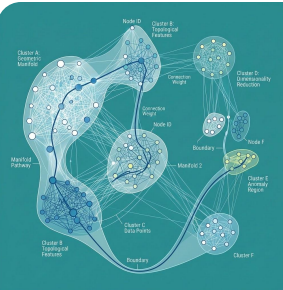
Preserves local neighborhood relations.

WHEN TO USE
2D/3D cluster visualization.

EXAMPLE
Single-cell RNA sequencing.

UMAP

GLOBAL STRUCTURE



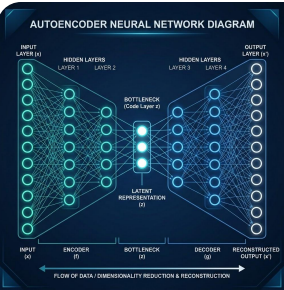
Preserves local & global topology.

WHEN TO USE
Scalable non-linear reduction.

EXAMPLE
Big data visualization.

Autoencoder

NEURAL BOTTLENECK



Compresses via neural bottleneck.

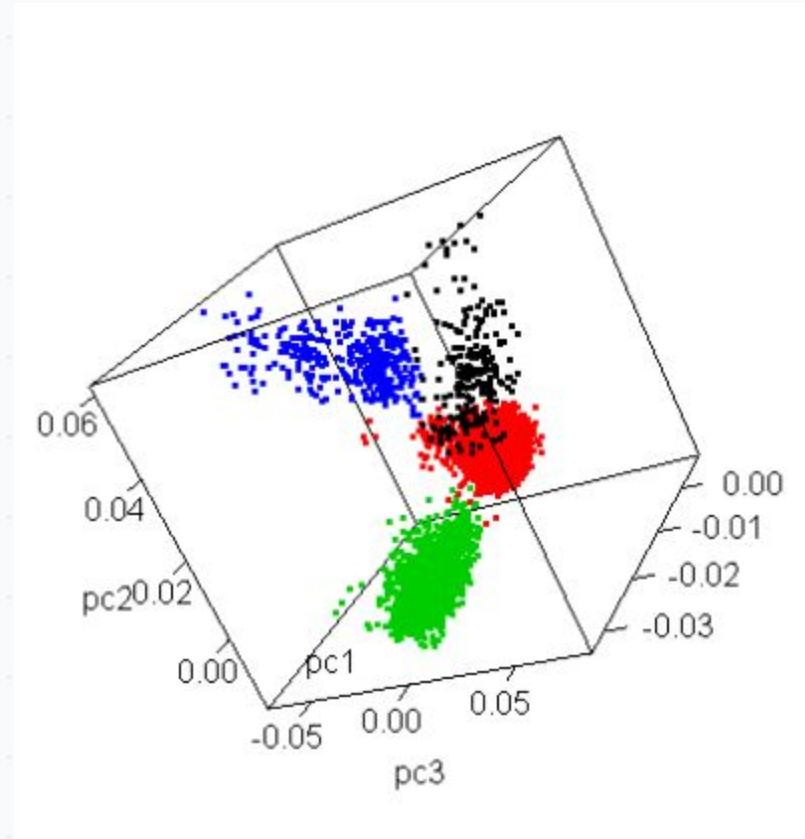
WHEN TO USE
Complex non-linear encoding.

EXAMPLE
Image denoising & anomaly detection.

Principal Component Analysis (PCA)

The Intuition: Finding Maximum Variance

- **The Problem:** Analyzing 20 variables describing cities (internet, cost, safety) is overwhelming. Which variables truly matter?
- **The Solution:** PCA finds new "composite variables" (Principal Components) ranked by importance.
- **How it works:** It combines multiple original variables into a few dimensions. Keep only the top few that capture the most variance.
- **Example:** 20 original variables collapse into PC1 ("Overall Quality of Life") and PC2 ("Cost vs. Amenities").



Variance creates room for information

Analyzing 100 cities dataset: Why variable differences unlock structural patterns

City Feature Matrix (N = 100)

City	Avg Temp (°C)	Rainy Days	Country
City 1	8	20	Denmark
City 2	10	18	Denmark
City 3	12	15	Denmark
City 4	25	5	Denmark
City 5	27	3	Denmark
...	...	...	...
City 100	15	12	Denmark

Key Observation: Population also varies drastically (0.8M to 5.2M).
Constant features fail to discriminate between observations.



Which feature gives
us the least
information for
distinguishing these
100 cities?

Join at
slido.com
#3459 159

Variance creates room for information

Analyzing 100 cities dataset: Why variable differences unlock structural patterns

City Feature Matrix (N = 100)

City	Avg Temp (°C)	Rainy Days	Country
City 1	8	20	Denmark
City 2	10	18	Denmark
City 3	12	15	Denmark
City 4	25	5	Denmark
City 5	27	3	Denmark
...	...	...	...
City 100	15	12	Denmark

Key Observation: Population also varies drastically (0.8M to 5.2M).
Constant features fail to discriminate between observations.

Variance & Information Analysis

HIGH VARIANCE

Temperature, Rain, Population
Substantial spread provides signal to uncover correlations.

ZERO VARIANCE

Country Column (All Denmark)
Constant value offers zero capability to distinguish cities.

Investigative Questions Unlocked by Variance:

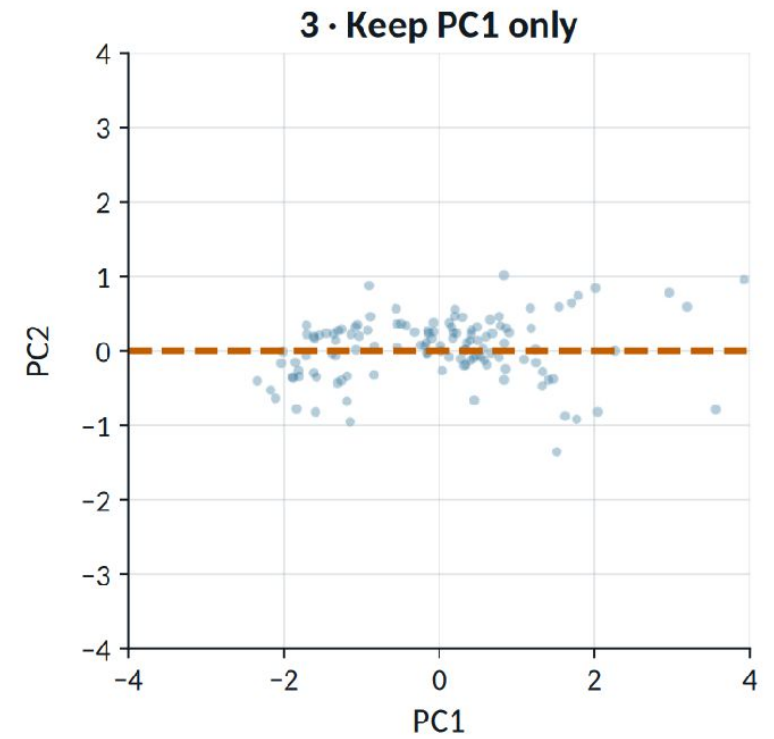
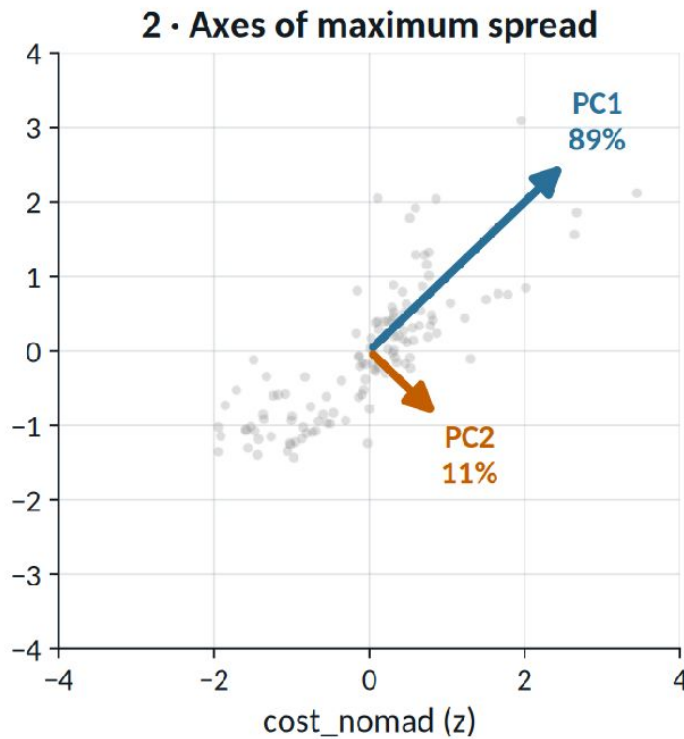
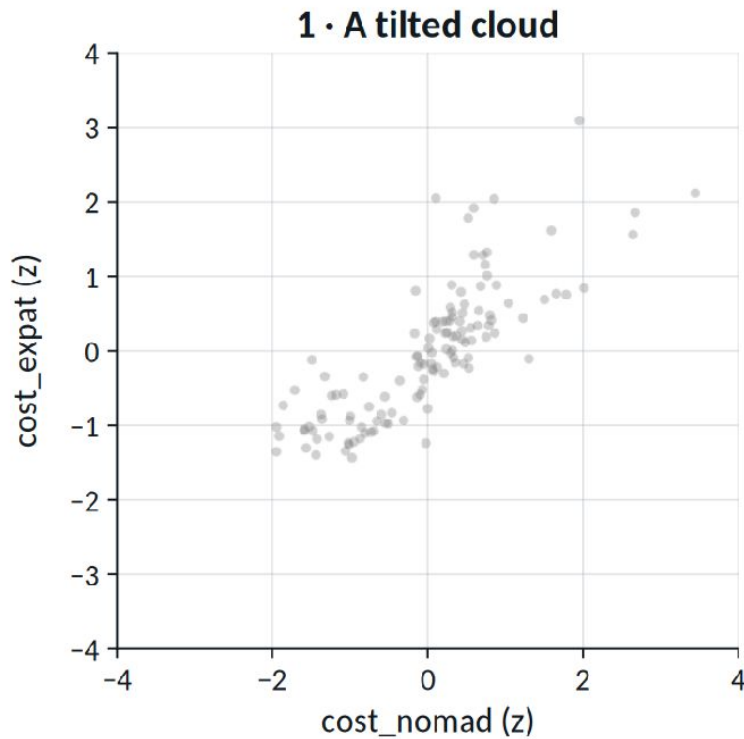
- **Weather Patterns:** Do cities with higher temp have fewer rainy days?
- **Urban Impact:** Do larger populations correlate with specific weather?
- **Dimensionality:** Which combined variables explain most spread?

Link to PCA:

PCA identifies axes of maximum variance. If a feature is constant, it yields no structure. High multi-variable variance allows PCA to build optimal lower-dimensional projections.

***Caveat:** High variance does not automatically guarantee useful signal—it can also stem from noise. Domain interpretation remains essential.*

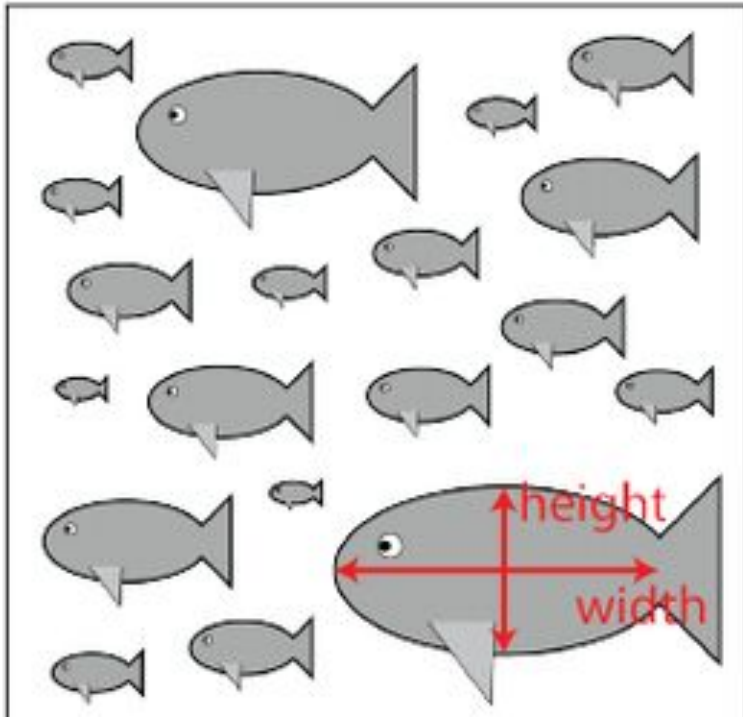
What PCA actually does



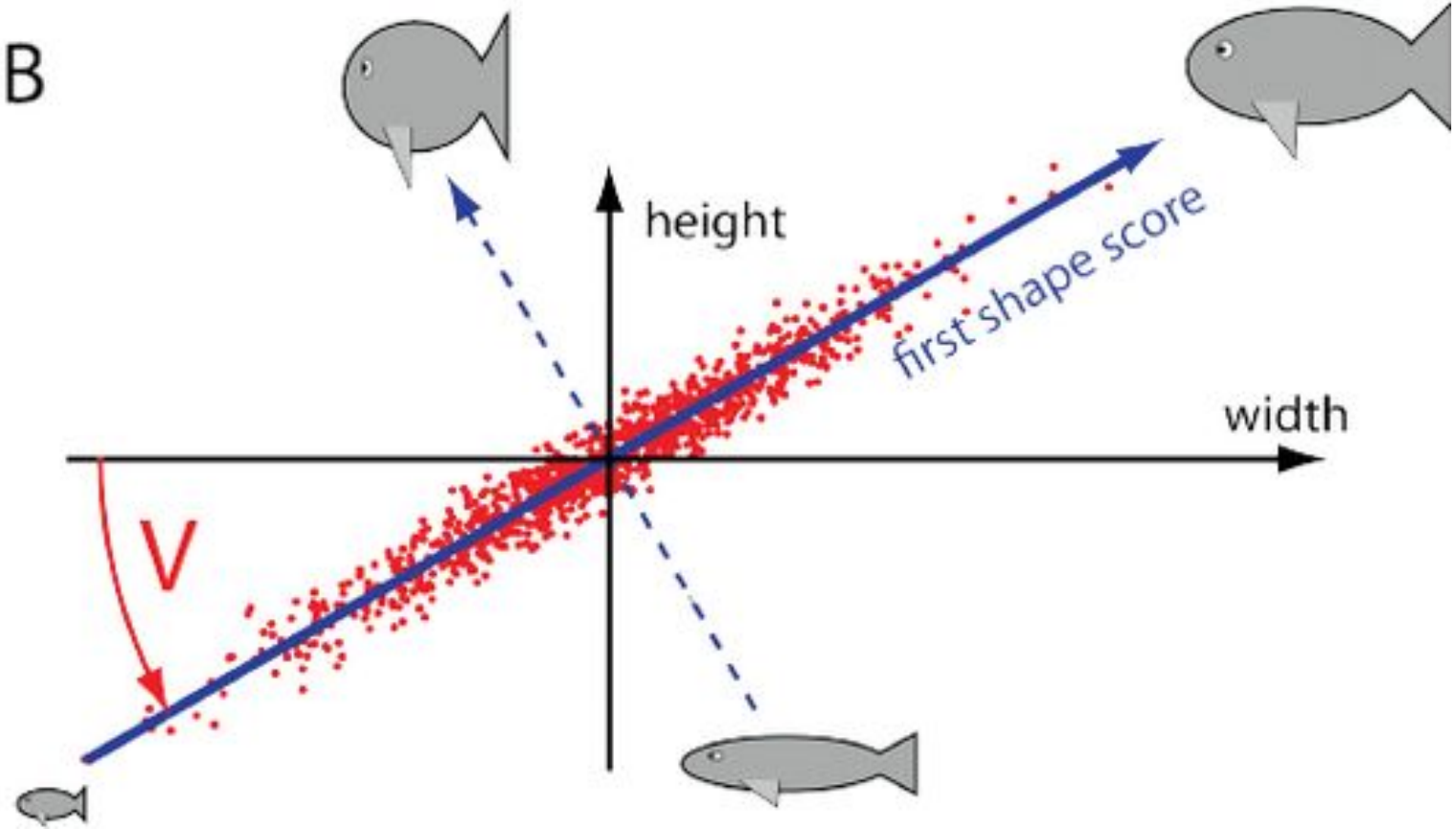
It TURNS THE AXES so the first one points along the direction the data spreads out most. Nothing is lost by turning. The reduction happens when you CHOOSE TO STOP after a few axes.

What PCA actually does

A



B



The toy Nomad dataset

Three cities, three positive features on a 0–10 scale

City	Nightlife	Cafés	Beach
Copenhagen	8.0	8.5	1.0
Lisbon	5.0	6.5	3.0
Bali	2.0	6.0	8.0

What we want to learn

Can we replace Nightlife, Cafés and Beach with fewer new dimensions?

For example:

- an “urban vs beach” dimension
- a general city-activity dimension

$X =$		Nightlife	Cafés	Beach
	Copenhagen	8	8.5	1
	Lisbon	5	6.5	3
	Bali	2	6	8

$$X = \begin{bmatrix} 8 & 8.5 & 1 \\ 5 & 6.5 & 3 \\ 2 & 6 & 8 \end{bmatrix}$$

The numbers are small on purpose: they make the mechanics visible.

Step 1 — Calculate variance & covariance

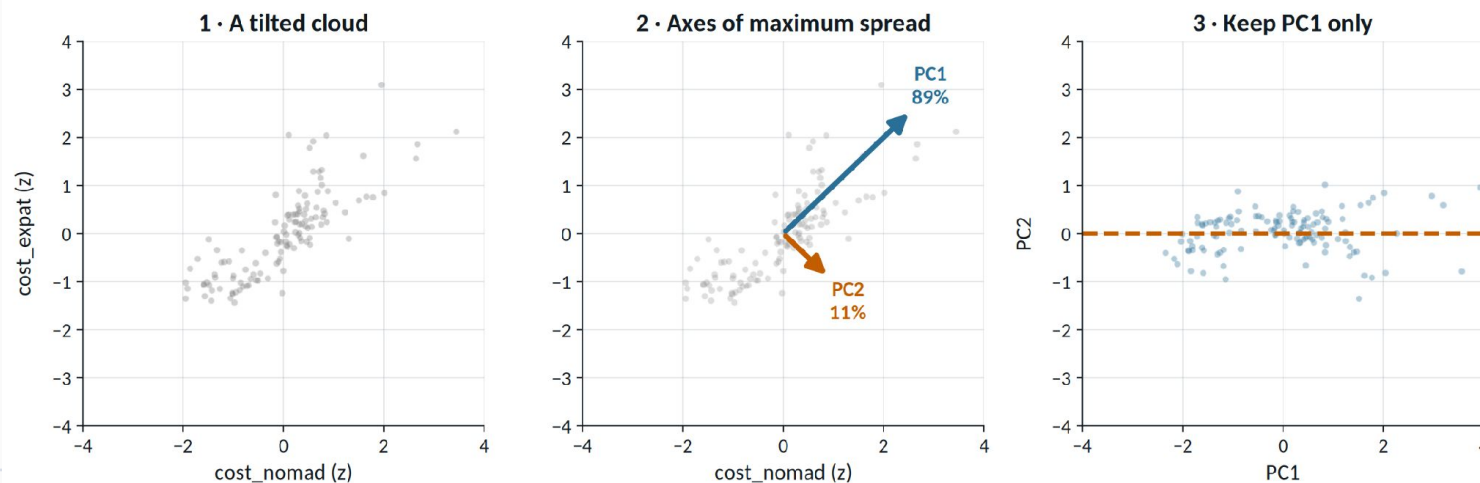
$\Sigma =$		<i>Nightlife</i>	<i>Cafés</i>	<i>Beach</i>
	<i>Nightlife</i>	9	3.75	-10.5
	<i>Cafés</i>	3.75	1.75	-4
	<i>Beach</i>	-10.5	-4	13

Step 2 — Calculate the PCA equation

$$PC1 = 0.620(\text{Nightlife}) + 0.248(\text{Cafés}) - 0.744(\text{Beach})$$

Urban lifestyle $\longleftrightarrow$ Beach lifestyle

A high PC1 score means **more urban**, while a low/negative score means **more beach-oriented**.



The whole session, in six steps

1

Load

Read the data. One line.

2

Clean

Fix the three things that are wrong with this file. One line each.

3

Standardise

Put every column on the same scale. Skip this and everything below is wrong.

4

Run PCA

Two lines. Genuinely the easiest part.

5

Read it

How many components to keep, and what they actually mean. The part that takes judgement.

6

Plot it

The picture you wanted in the first place.

Then twelve minutes on t-SNE and UMAP, and a hands-on lab.

Step 1 · The 19 Features (Dataset Metrics)

Overview of variables analyzed across 125 cities before applying PCA dimensionality reduction.

#	Column Name	Plain Words Description	Typical Range
1	cost_nomad	Monthly cost of living for a remote worker, in dollars	780 – 6,150
2	cost_coworking	Monthly price of a desk in a coworking space	15 – 1,505
3	cost_expats	Monthly cost of living for a foreigner living there	690 – 7,890
4	coffee_in_cafe	Price of a coffee in a café, in dollars	0 – 10
5	places_to_work	How easy it is to find somewhere to work from	0 – 1
6	free_wifi_available	How widely free Wi-Fi is available in public	0 – 1
7	internet_speed	Average internet speed, in Mbps	0 – 197
8	freedom_score	How free the country is (civil and political rights)	0 – 19
9	peace_score	How peaceful the country is (conflict, crime, unrest)	0 – 1
10	safety	How safe the city feels day to day	0 – 1
11	fragile_states_index	How close the state is to breaking down — high = worse	0 – 79
12	press_freedom_index	Press-freedom score — high = worse for journalists	9 – 45
13	female_friendly	How comfortable the city is for women	0 – 1
14	lgbt_friendly	How welcoming the city is to LGBT people	0 – 1
15	friendly_to_foreigners	How welcoming locals are to outsiders	0 – 1
16	racism	How much racism visitors report — high = worse	0 – 1
17	leisure	How much there is to do — culture, sport, outdoors	0 – 1
18	nightlife	How good the bars, clubs and evening scene are	0 – 1
19	weed	How tolerated cannabis is	0 – 1

These 19 continuous features represent economic, social, infrastructure, and freedom dimensions across 125 target global cities.

The data: cities to live and work in

One row per city. Each column is one thing somebody measured about it.

Cost

5

rent for a nomad, for an expat, coworking desk, a coffee, a beer

Work

3

places to work from, free wifi, internet speed

Safety & institutions

5

freedom, peace, safety, state fragility, press freedom

Lifestyle

7

how welcoming, how much to do, nightlife

life_score

1

the website's own overall rating — we set this one aside

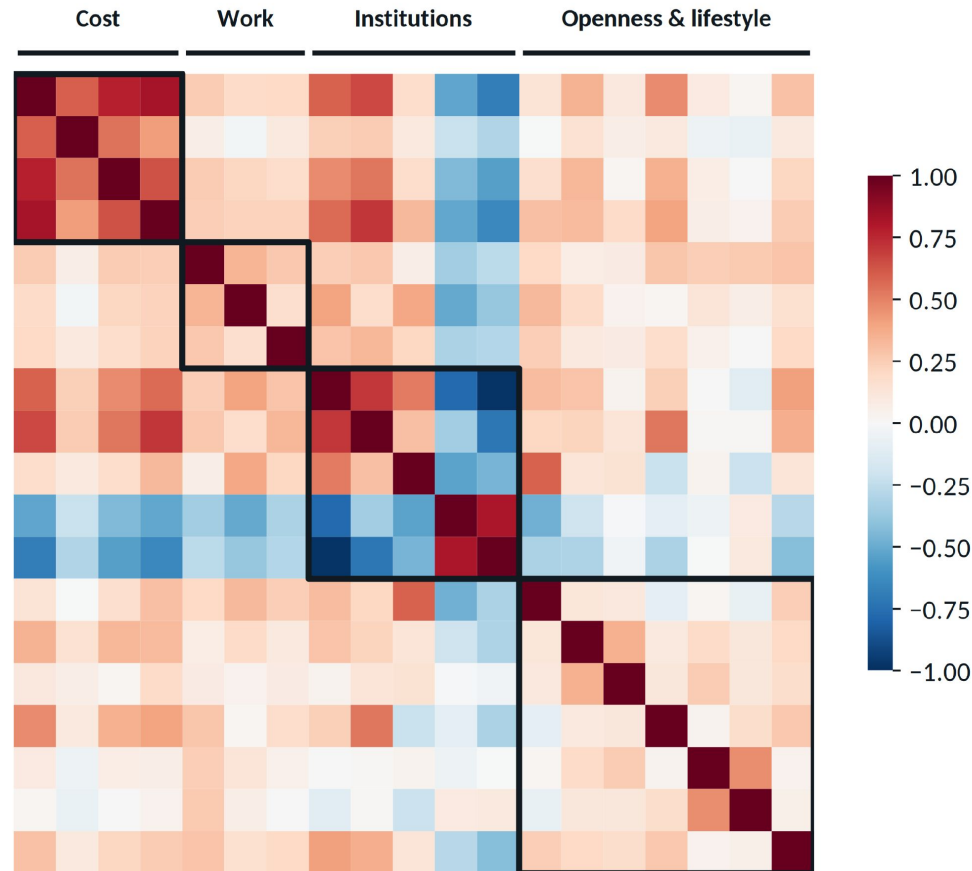
Labels

4

city, country, region — never given to any method

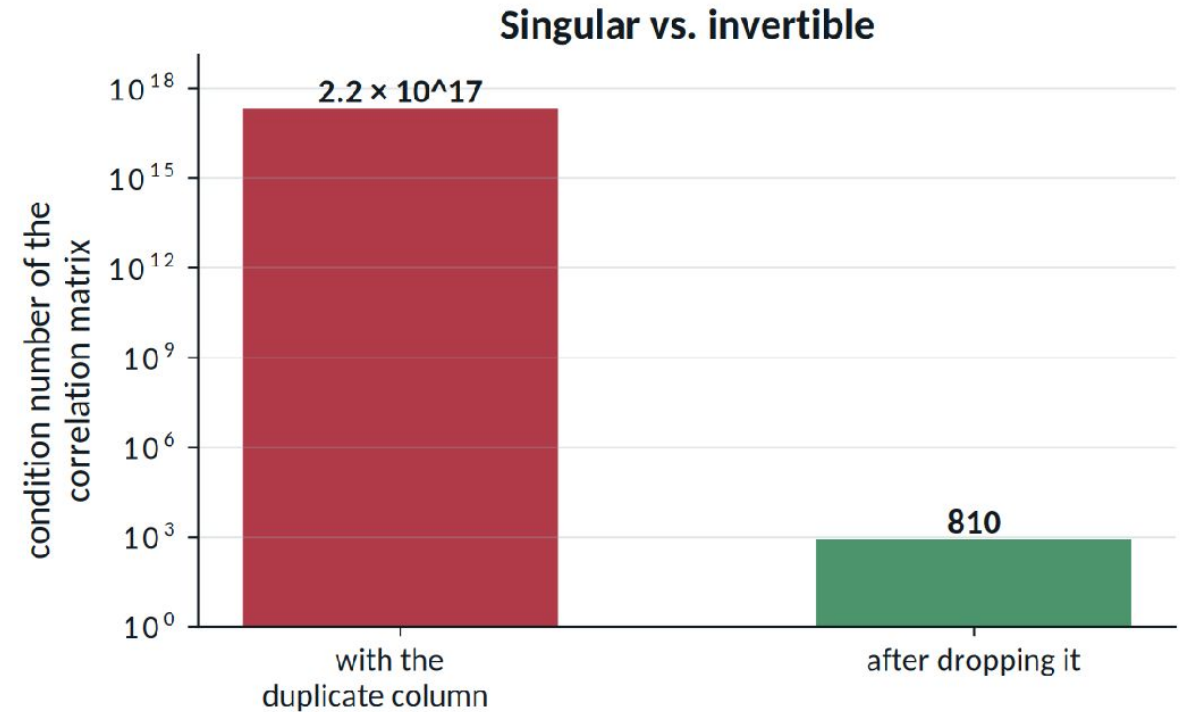
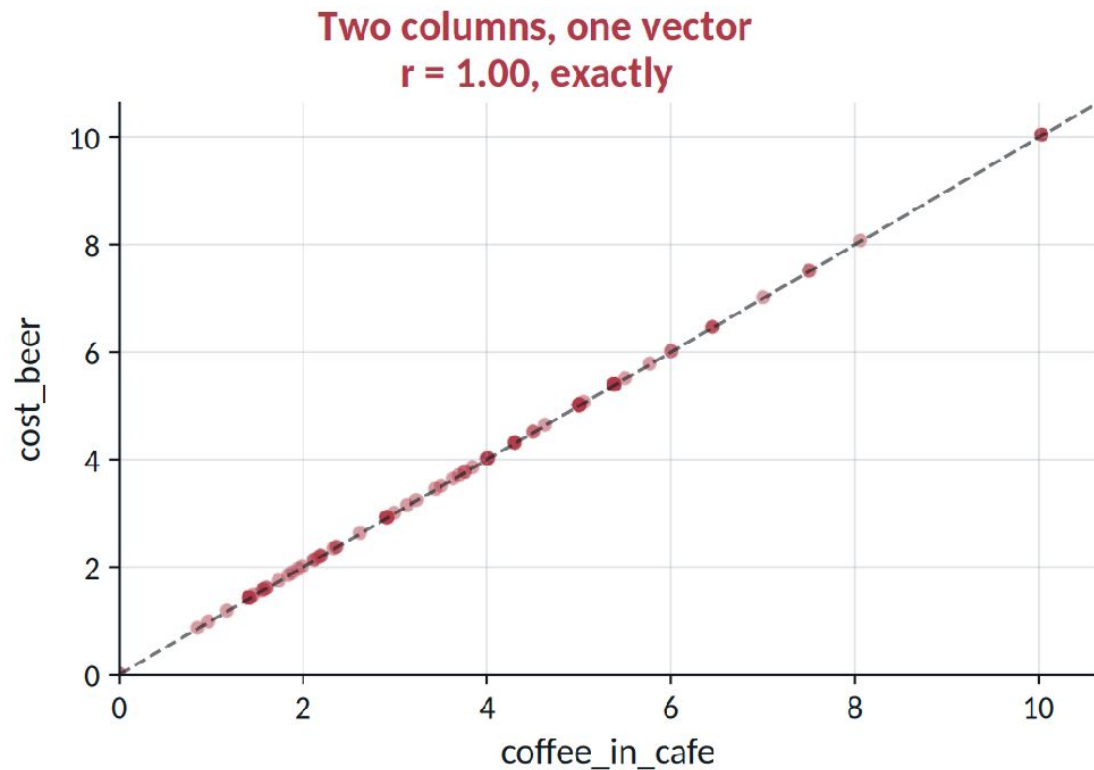
Real, public data — which means it is NOT clean. That is step 2.

Why bother? The columns repeat each other



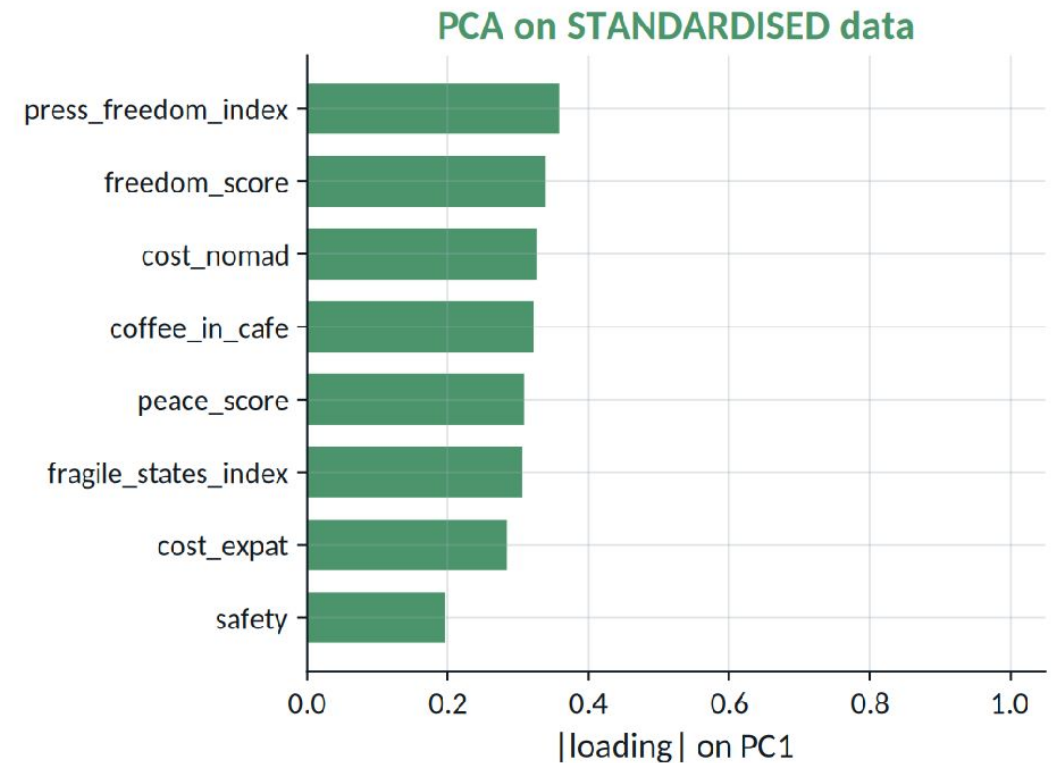
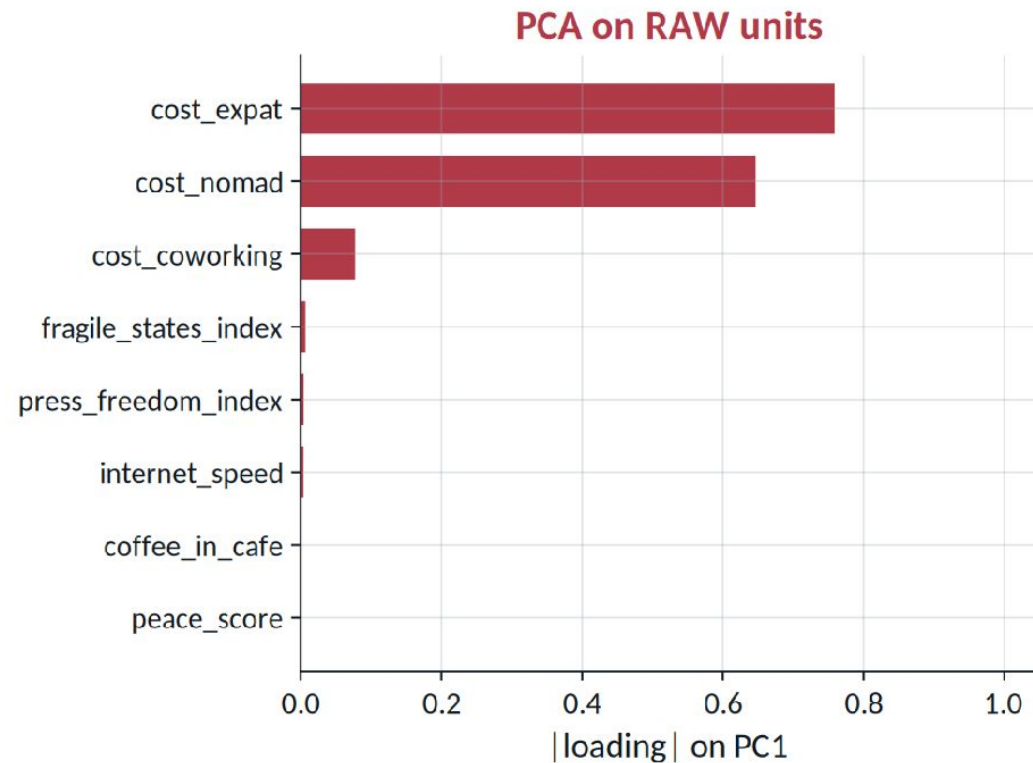
Dark squares = two columns that move together. The cost columns clump. The safety columns clump. 19 columns, but nowhere near 19 separate pieces of information — and to see every pair by eye you would need 171 scatter plots.

Step 2 · Clean it — three one-line fixes



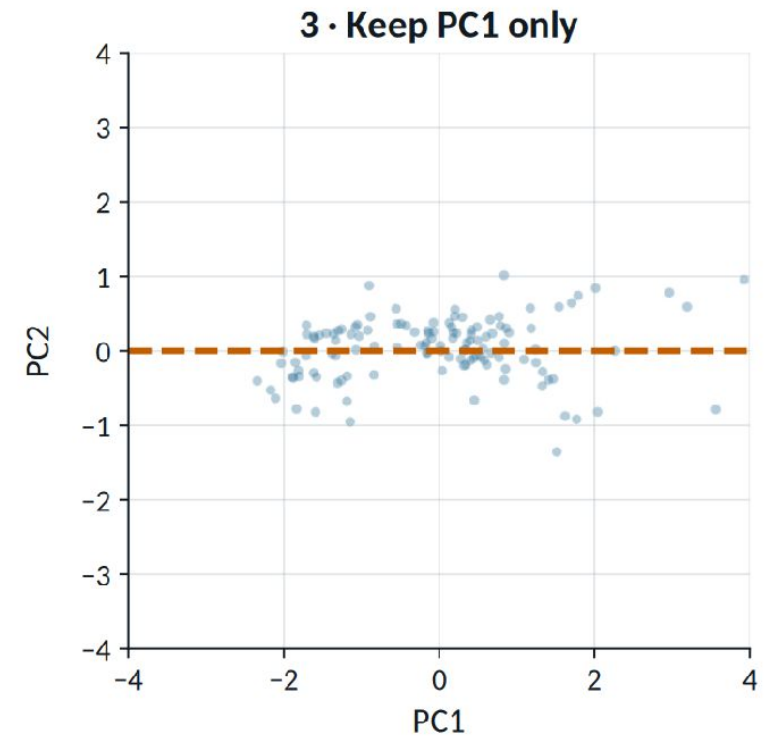
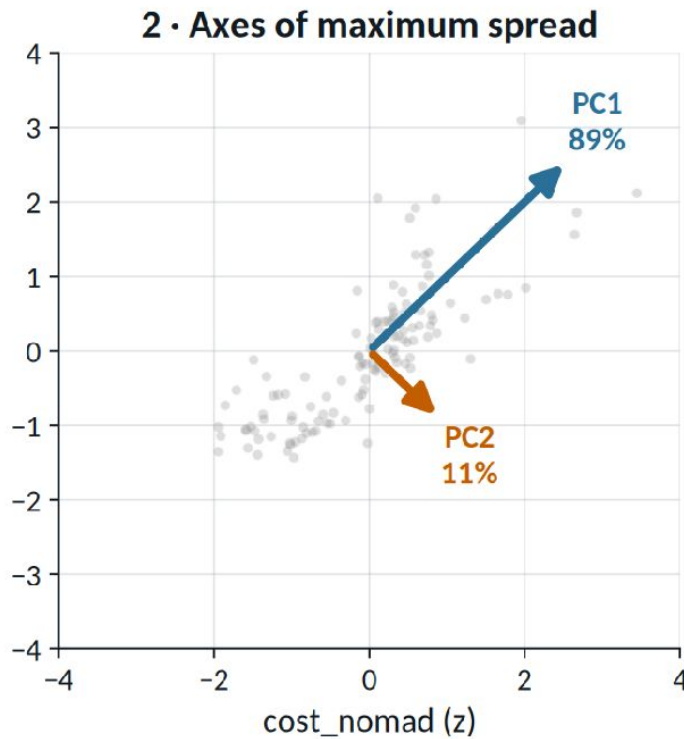
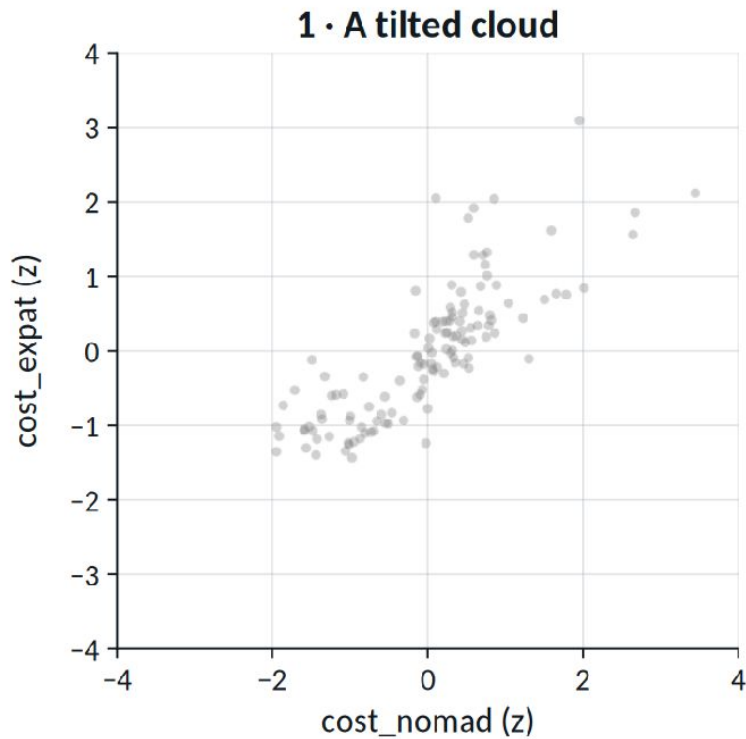
1. coffee_in_cafe and cost_beer are identical in every row — drop one.
2. Two columns contain $5.75e-30$ and $-3.51e-17$ where zero belongs — round them.
3. life_score is calculated FROM the other columns — set it aside.

Step 3 · Standardise — the step people skip



Rent is in THOUSANDS of dollars. Safety is between 0 and 1. PCA chases whatever varies most — so on raw numbers it decides cost_expats is 88.2% of everything. That is not a finding about cities. It is a finding about dollars.

Step 4 · What PCA actually does



It TURNS THE AXES so the first one points along the direction the data spreads out most. Nothing is lost by turning. The reduction happens when you CHOOSE TO STOP after a few axes.

Step 4 • The code

```
# standardise
scaler = StandardScaler()
X = scaler.fit_transform(features)

# run PCA
pca = PCA(n_components=5, random_state=42)
scores = pca.fit_transform(X)
```

That is the whole thing. scores is your new table: same cities, 5 columns instead of 19.

What you get back

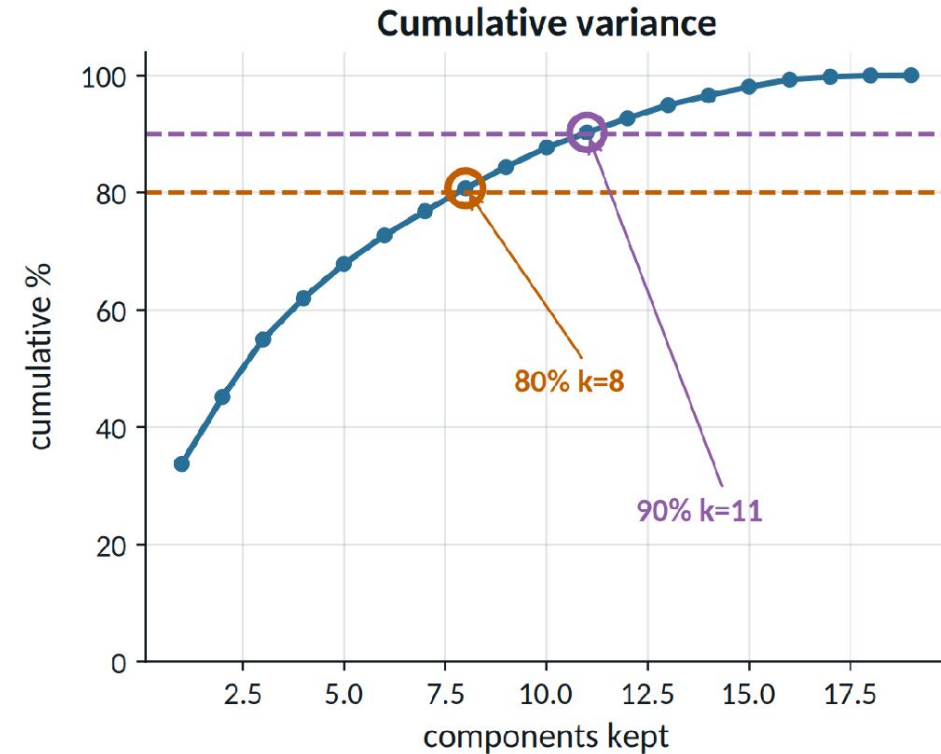
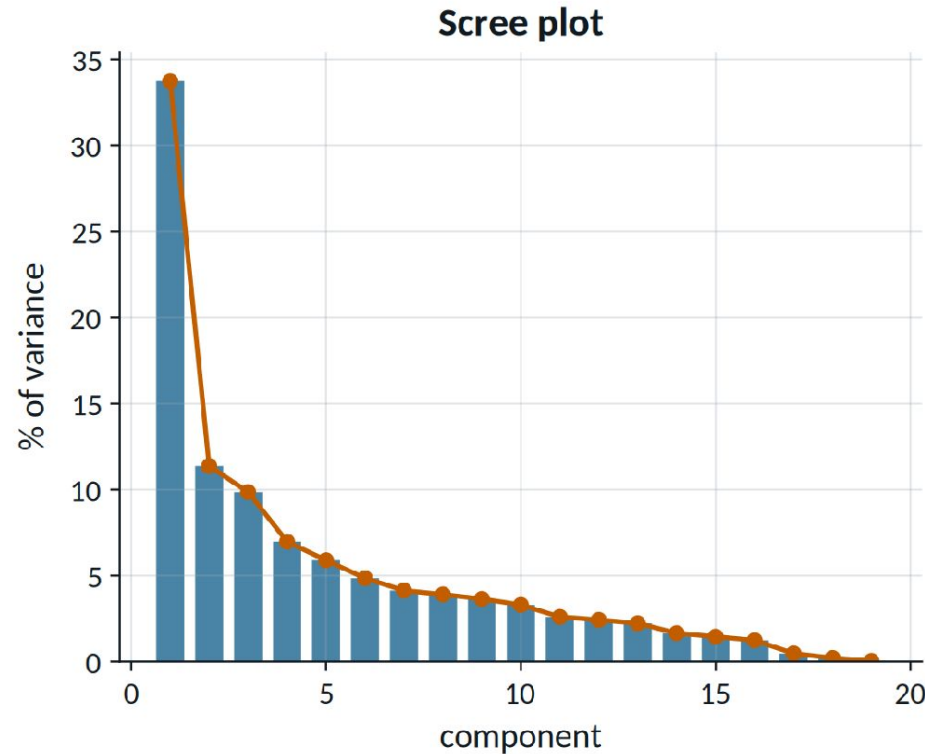
explained_variance_ratio_ — how much each component captured. components_ — the LOADINGS: which columns make up each component. scores — where each city sits on the new axes.

Two words to keep straight

LOADINGS are about columns. SCORES are about rows.

19 columns → 5. The next two slides are about deciding whether 5 was right, and what those 5 actually are.

Step 5a · How many components should you keep?



33.8%

PC1 alone

11.4%

PC2

9.9%

PC3

8

to reach 80%

2

if you only want a picture

Three rules: the ELBOW in the left plot · the 80% LINE in the right plot · JUST 2 if you only want to plot it. Pick one — and say in your report which one you picked.

Step 5b · What do the components mean?

The loadings table is hard to read. SORT ONE COMPONENT AT A TIME — this is the whole trick.

```
loadings["PC1"].sort_values()
```

The columns at the TOP (big negative) and the BOTTOM (big positive) are the two ends of ONE AXIS. A city with a high PC1 score is high on the bottom ones and low on the top ones.

PC1 on our data

One end: freedom_score, cost_nomad, coffee_in_cafe. Other end: press_freedom_index, fragile_states_index.

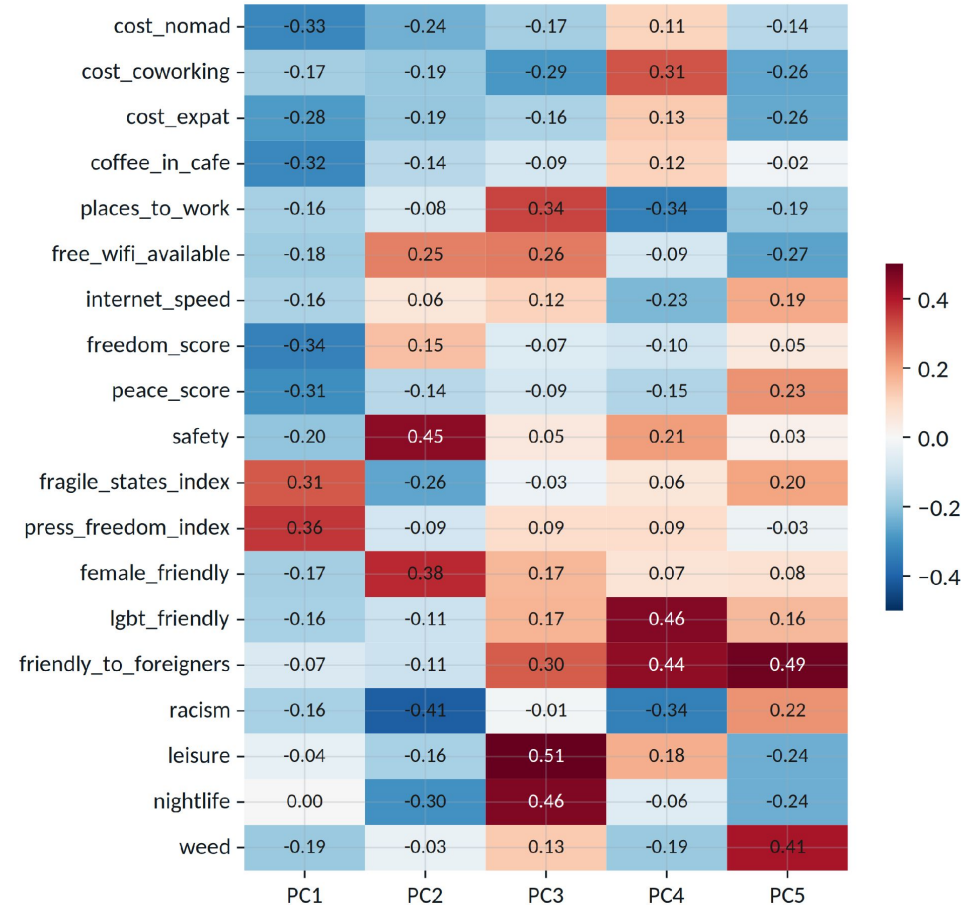


Plus and minus are arbitrary

PCA could equally have returned every number with the sign flipped — same axis, pointing the other way. So never say “PC1 is positive, therefore good”. Only the CONTRAST means anything.

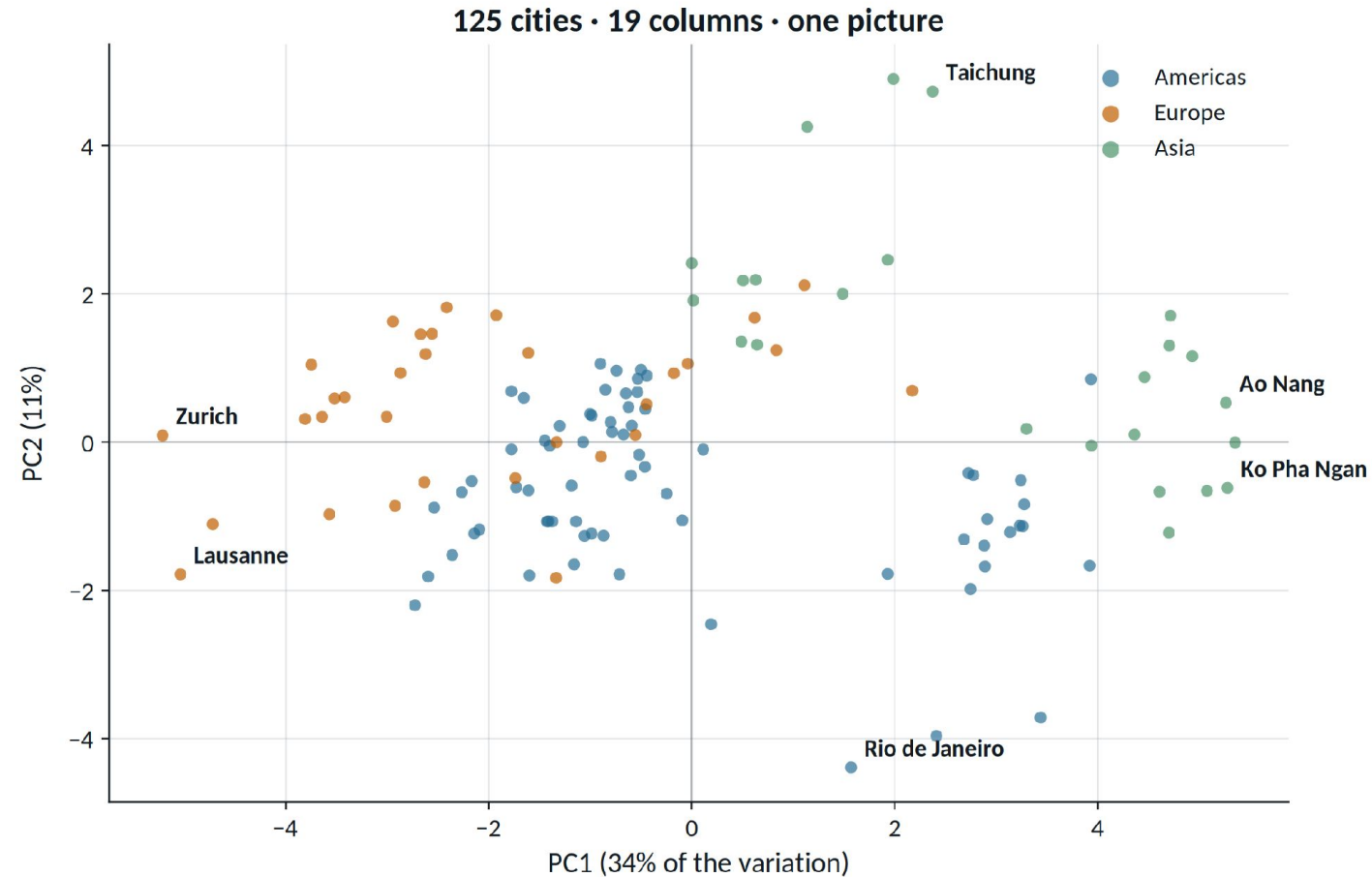
So PC1 is not “one variable”. It is a contrast between two groups of variables — and your job is to give that contrast a name.

The same thing as a picture



Red = the column pushes that component UP. Blue = it pushes it DOWN. White = that column has nothing to do with this component. Read each COLUMN of this heatmap as one axis.

Step 6 · The picture



125 cities, 19 columns, one plot. And we NEVER TOLD IT ABOUT THE REGIONS — the colours were added afterwards. That separation is structure PCA found on its own.

WHICH CITIES SIT AT EACH END OF PC1?

And does the name you gave it hold up?

Zurich

one end — freedom_score, cost_nomad

Lausanne

expensive, high freedom and peace scores

Ko Tao

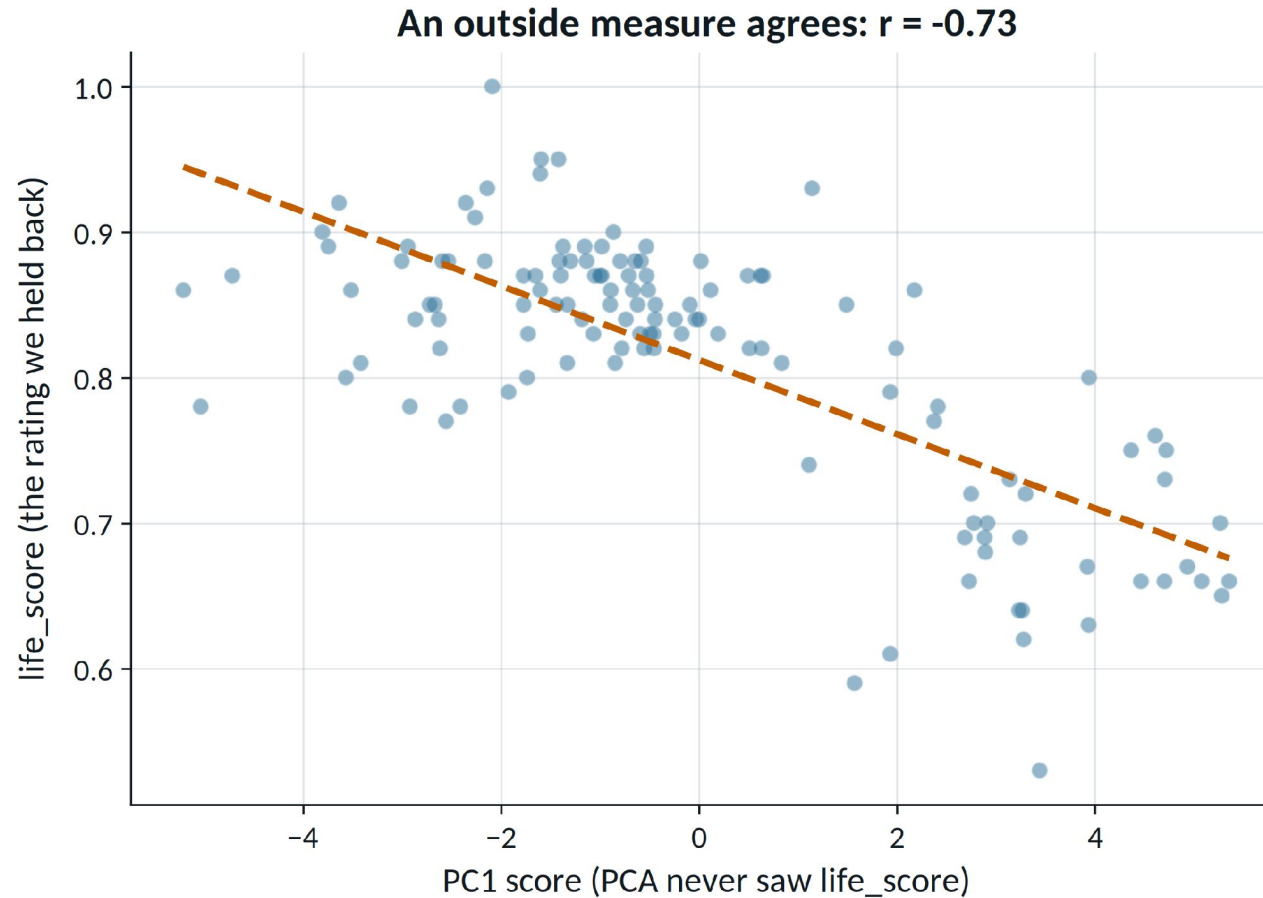
the other end — press_freedom_index,
fragile_states_index

Hua Hin

cheap, more fragile institutions

THIS IS THE TEST. Sort your cities by the component and look at both ends. If the names match the label you gave it, you have a finding. If they do not, you named it wrong — go back to the loadings.

A free check: does it agree with something we hid?



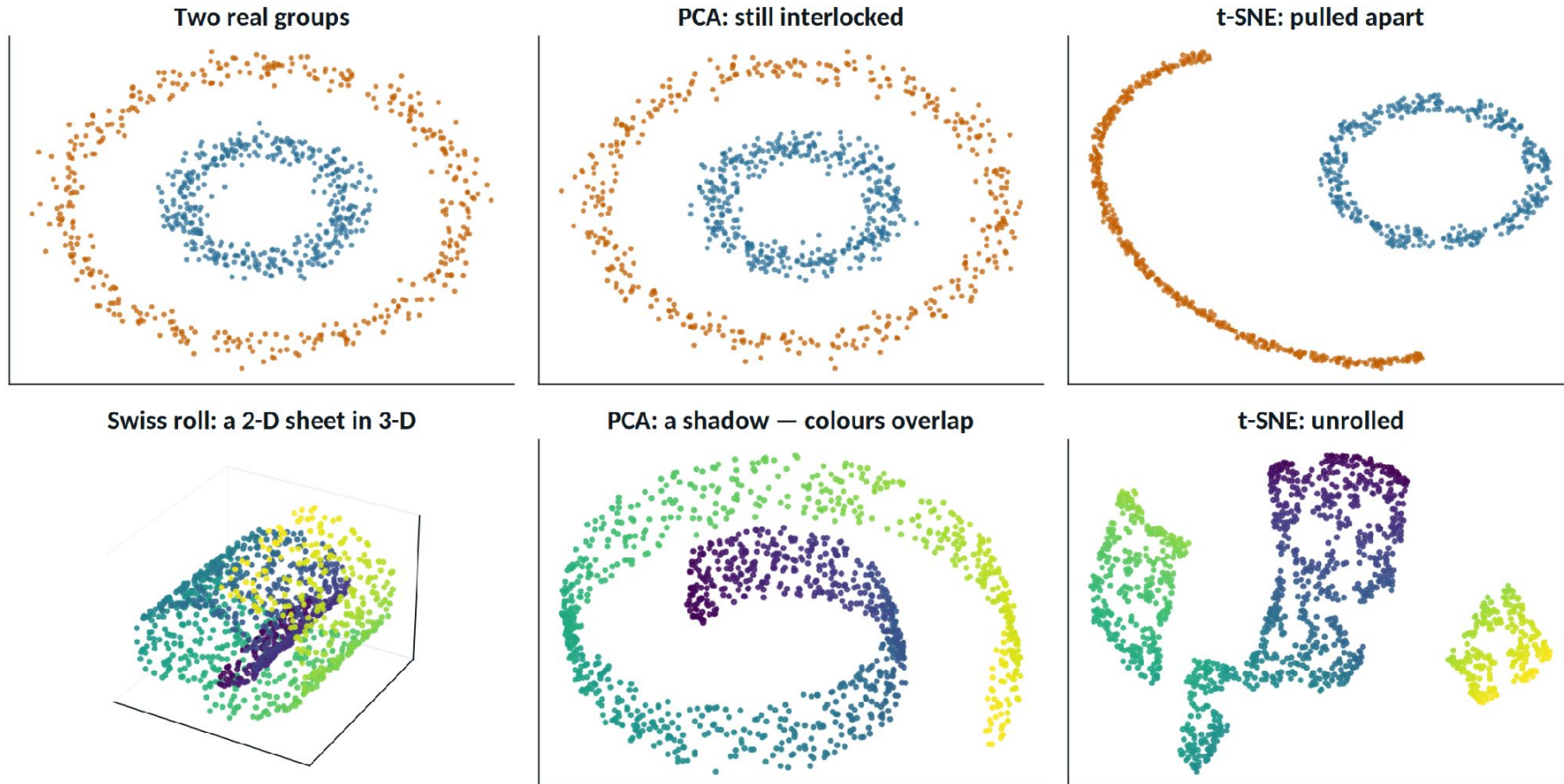
Remember we set `life_score` aside in step 2. PCA has NEVER SEEN IT — and PC1 lines up with it at $r = -0.73$. An outside measure agrees with what PCA found on its own. **High PC1 score means a cheap, less-free, more-fragile place.**

2

t-SNE and UMAP

Two other ways to get a 2-D picture —
and how to not be fooled by them.

PCA can only draw straight lines



Top row: two real groups, one inside the other. PCA can turn the picture but never separate them. t-SNE can. Bottom row: a flat sheet rolled up — PCA casts a shadow where far-apart points land on top of each other.

What they do — and what they don't

Both methods do the same job: place every row on a 2-D map so that SIMILAR ROWS END UP NEAR EACH OTHER.

```
tsne = TSNE(n_components=2,  
            perplexity=30,  
            random_state=42)  
coords = tsne.fit_transform(X)
```

UMAP: same idea, faster — and unlike t-SNE it can place NEW rows onto an existing map later. That is what makes it usable in a real system.

What they optimise

Who is near whom. That is all.

What they do NOT

The DISTANCE between clusters. The SIZE of clusters. The DIRECTION of the axes. None of it is optimised, so none of it carries meaning.

Every mistake people make with these plots comes from reading the second box as if it were the first.

When should we use PCA, t-SNE / UMAP, or both?

Use the method that matches your goal: understanding, showing, or both.

USE PCA

When you want to understand the main dimensions in the data

Reduce many variables into a few interpretable components

Good for loadings, variance explained, and model inputs

Example: 30 survey items → 3 business factors

USE t-SNE / UMAP

When you want to see hidden groups or curved structure

Best for 2D visualization and pattern exploration

Good for clusters and local neighborhoods

Example: show customer segments visually

USE BOTH

When you want interpretation first and visualization second

Use PCA to reduce noise and summarize structure

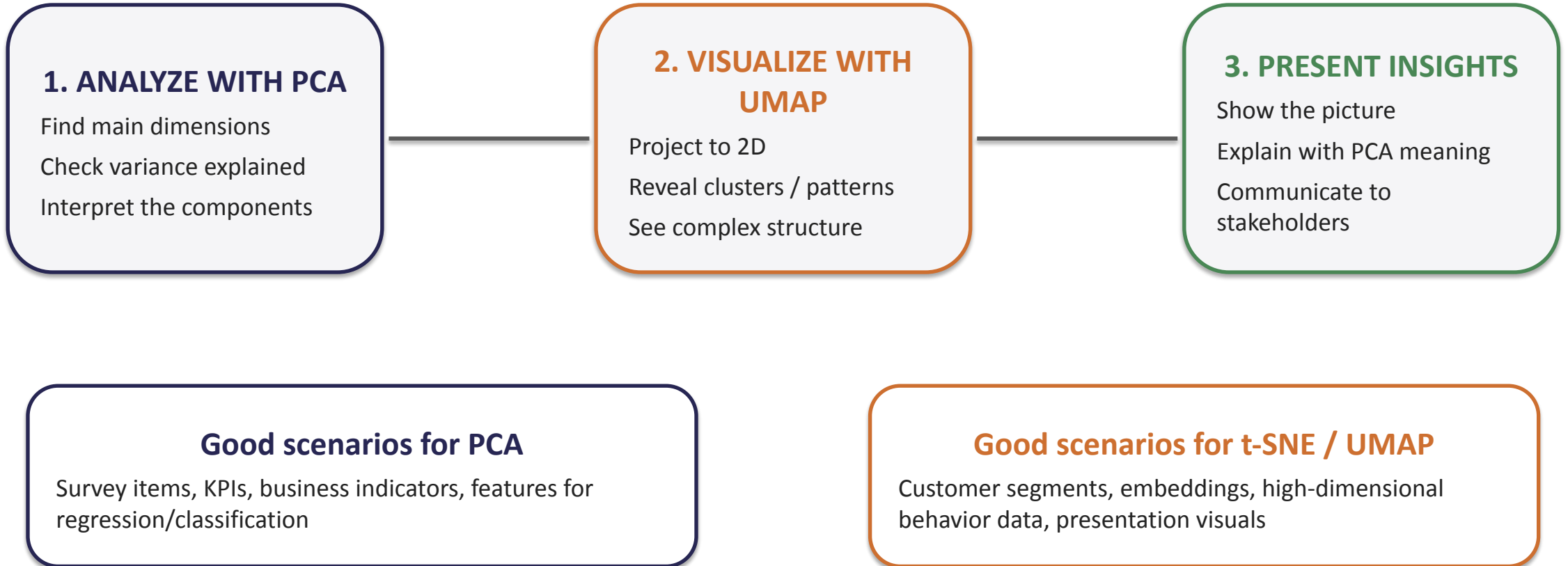
Then use UMAP or t-SNE to create a clear 2D picture

Example: 100 variables → PCA(10) → UMAP(2)

Simple rule: PCA = understand | t-SNE / UMAP = show | PCA + UMAP = understand first, then show

Recommended workflow

A simple classroom workflow for using both linear and non-linear methods.



Teaching shortcut: Use linear methods for understanding, non-linear methods for showing.

Four rules for reading these plots

1

The distance between two blobs means nothing. Two groups far apart are not more different than two groups close together. Do not build a story on the gaps.

2

The size of a blob means nothing. These methods spread out sparse regions and squash dense ones on purpose. A big blob is not a big group.

3

The axes mean nothing. There is no “tSNE1 = cost”. Never label them, never put them in a model.

4

Random noise makes convincing clusters. You will prove it yourself in Exercise 4.

A cluster in one of these plots is a QUESTION TO GO AND CHECK, never an answer.

Questions & Key Takeaways

01. Main Dimensions

PCA identifies fundamental linear components and variance, helping reduce complex datasets to key interpretable factors.

02. Visual Patterns

t-SNE & UMAP reveal non-linear clusters and hidden structures in 2D space for visual pattern exploration.

03. Combined Workflow

Use linear methods first to understand structure and reduce noise, then non-linear projection to communicate insights.